Work Interruptions and the Child Penalty: The role of parental gender*

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Abstract

This paper examines within-day work interruptions due to childcare as a candidate micro-level mechanism behind the child penalty. Using time-use diaries from the American Time Use Survey (ATUS, 2003–2023), we apply survival analysis combined with Random Survival Forests, Gradient Boosting, SHAP values, and Accumulated Local Effects plots to model the time until a working parent's first interruption. Parental gender is the dominant predictor across all methods: fathers face an instantaneous interruption hazard about 61 per cent below mothers' (HR = 0.389; 95% CI: 0.346–0.438), a gap robust to competing-risks decomposition, alternative censoring thresholds, and a stratified specification. Higher education and income are associated with greater, not lower, interruption risk. A cross-national extension to eight countries confirms the universality of the gender gap, with male hazard ratios below one everywhere (0.388 in the United States to 0.657 in Finland). These findings document a high-frequency margin of gender inequality and provide cross-sectional evidence of a daily mechanism that may contribute to explaining the child penalty.

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1 Introduction

The convergence between men's and women's labour-market outcomes has stalled, and the leading explanation for what remains is a phenomenon we observe only at low frequency. Since the 1990s the male–female earnings ratio has plateaued at around 80–82 per cent in the United States (Blau & Kahn, 2017; Olivetti & Petrongolo, 2017), and the *child penalty* — the sharp, persistent fall in mothers' but not fathers' earnings, hours, and participation after childbirth — accounts for the majority of the gap in high-income economies (Cortés & Pan, 2023; Kleven, Landais, & Leite-Mariante, 2025; Kleven, Landais, & Søgaard, 2019). Yet this penalty is measured in years, while the conflict between paid work and childcare is fought minute by minute within the workday, a scale at which it has never been observed. We therefore ask: *what predicts and modulates the risk of short-term, within-day work interruptions due to childcare?*

Theory suggests that the daily margin should matter. Modern accounts locate the penalty not in skill depreciation but in a labour market that rewards uninterrupted availability — Goldin's (2014) "greedy jobs" — combined with a form of care that is continuous and hard to delegate: the "on-call" supervision that Cuevas-Ruiz et al. (2025) show remains roughly constant as children age. Evidence comparing biological, adoptive, and same-sex couples largely rules out childbirth and fathers' labour-market advantage, leaving gender norms and preferences as the operative forces (Andresen & Nix, 2022; Bertrand, Kamenica, & Pan, 2015; Kleven, Landais, & Søgaard, 2021). If what employers reward is the absence of interruption, then the within-day episodes in which caregiving pulls a parent away from work are precisely where these forces should bite.

Observing that margin requires data that preserve the order of activities across the day rather than daily or hourly totals. Time-use diaries provide exactly this — a minute-level record of sequential activities — yet existing time-use research typically aggregates into hourly blocks (Giménez-Nadal & Molina, 2022; Giménez-Nadal, Molina, & Velilla, 2020), and survival analysis has been reserved for long-duration transitions such as spells of unemployment, obscuring the high-frequency transitions within the workday. We close this gap by adapting social sequence analysis (Abbott, 1995; Cornwell, 2015) to represent each diary as an ordered sequence of states and to detect the pattern work → childcare → work. This makes an event that is theoretically central, but empirically invisible at coarser resolution, directly measurable.

We bring this approach to the American Time Use Survey (ATUS, 2003–2023) and extend it to eight countries in the Multinational Time Use Study (MTUS). The US analytical sample comprises 36,339 working parents (1,544 interruption events) and the pooled multi-country sample 42,344 parents (1,915 events). An interruption is an episode of childcare that begins after the workday has started and is followed by a return to paid work; we further distinguish *pure* interruptions (the interval is exclusively childcare) from *mixed* ones (childcare combined with leisure or personal care). Because the design conditions on having worked at least one hour on the diary day, it characterises the *intensive* within-day margin rather than the extensive participation margin — a scope restriction we make explicit throughout.

We model the time elapsed until the first interruption. Cox proportional-hazards models with survey weights and household-clustered standard errors form the backbone, with a

stratified specification to address the detected violation of proportional hazards and a competing-risks layer (cause-specific and Fine–Gray) that exploits the pure/mixed typology; Random Survival Forests, Gradient Boosting, SHAP values, and Accumulated Local Effects plots provide a flexible cross-check on functional form and on the ranking of predictors. Casting the problem as "time until interruption" rather than a binary outcome lets the data speak to *when*, not just *whether*, caregiving interrupts paid work, while the machine-learning layer guards against imposing the wrong functional form.

Across every method, parental gender is the dominant predictor of interruption risk. Fathers face an instantaneous hazard 61 per cent below mothers' (HR = 0.389; 95% CI: 0.346–0.438; $p < 2 \times 10^{-16}$), an estimate that is essentially unchanged across the stratified specification, alternative censoring thresholds, and the competing-risks models (pure: 0.383; mixed: 0.420), and that dwarfs the role of income. Alongside this gap, higher education and income predict *more*, not fewer, interruptions — a pattern consistent with intensive parenting norms and the short-run autonomy that "greedy jobs" afford to higher-earning parents (Goldin, 2014; Guryan, Hurst, & Kearney, 2008). The gender gap is universal across the eight countries, with male hazard ratios below one everywhere (roughly 0.39 in the United States and France to 0.66 in Finland). That gender outweighs economic status points away from a purely resource-based account and toward one rooted in who is expected to respond to children during the workday.

Three threats to interpretation deserve emphasis, and we confront each. The design is observational, so we speak of associations rather than causal effects; it conditions on working that day, leaving the extensive margin — the larger part of the long-run penalty — outside our frame; and interruptions may be more visible in the diaries of teleworkers and autonomous professionals, which could inflate the privilege gradient. We address the statistical concerns directly (stratification for non-proportional hazards, household clustering, sensitivity to the censoring threshold, and competing risks), while flagging the selection and visibility issues as genuine bounds on what the estimates can claim. These caveats narrow the interpretation without overturning the central finding, whose magnitude and direction survive every robustness check.

Two features further guard against the gap being an artefact. First, it replicates in all eight national contexts, so it cannot be a product of the ATUS sampling frame alone. Second, methods with very different assumptions agree: the semi-parametric Cox, gradient boosting, and the forest deliver similar discrimination (C-indices of 0.72, 0.72, and 0.60), and the permutation-VIMP, SHAP, and Cox rankings all place gender and child age at the top while revealing the same non-linearities. When independent models and independent countries point the same way, the case that a single specification or context is driving the result becomes hard to sustain.

Our main contribution is to show, using time-use diary data, that a large gender gap in within-day work interruptions due to childcare exists across multiple national contexts. This cross-sectional evidence is relevant to the child penalty literature (Cortés & Pan, 2023; Kleven, Landais, & Søgaard, 2019): it documents a daily mechanism that is consistent with the penalty, and the cross-national replication provides a form of external validity for the mechanisms proposed in that literature. This complements quasi-experimental studies (Andresen & Nix, 2022; Kleven, Landais, & Søgaard, 2021) and evidence on the motherhood wage gap (Amuedo-Dorantes & Kimmel, 2005) by characterising the intensive, within-day

margin at a frequency those designs cannot reach. We do not provide causal evidence, nor do we trace how daily disruptions cumulate into long-run outcomes; that link is left for future work. Methodologically, we bridge intraday time-use data and event-history analysis, and we make the pure/mixed typology do empirical work by treating interruption types as competing risks.

The paper is structured as follows. Section 2 reviews the literature. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 discusses implications. Section 7 concludes.

2 Literature Review

2.1 The Child Penalty and the Mechanisms Behind It

Childbirth, rather than skill or schooling, is the principal inflection point in the persistence of gender inequality in the labour market. Using Danish administrative data, Kleven, Landais, and Søgaard (2019) estimate a long-run motherhood earnings penalty of roughly 20 per cent that opens at the first birth and never closes; the pattern recurs across institutional settings (Kleven, Landais, & Leite-Mariante, 2025), and in the United States children account for close to two-thirds of the gender earnings gap (Cortés & Pan, 2023). This evidence is difficult to reconcile with the human-capital view that wage gaps reflect skill depreciation during absences (Mincer, 1974), and it reframes the penalty as a structural feature of how work and care are organised — the premise that motivates our search for its daily mechanics.

The asymmetry between mothers and fathers arises from social norms interacting with a labour market that penalises inflexibility, not from biology. Comparing biological, adoptive, and same-sex couples, Andresen and Nix (2022) largely rule out childbirth itself and fathers' labour-market advantage, leaving gender norms and preferences as the operative forces — consistent with the adoptive-family evidence of Kleven, Landais, and Søgaard (2021) and with documented aversion to wives out-earning husbands (Bertrand, Kamenica, & Pan, 2015) and the intergenerational transmission of norms (Fernández & Fogli, 2009). On the demand side, Goldin (2014) shows that "greedy jobs" reward long, uninterrupted hours non-linearly, and Adda, Dustmann, and Stevens (2017) trace part of the penalty to the occupations women choose in anticipation of motherhood. Together these mechanisms imply that the binding friction is the parent's availability *within* the workday — precisely the margin we measure.

A complementary strand suggests that the burden of care is both rising and concentrated among advantaged families, foreshadowing a central result of this paper. Cuevas-Ruiz et al. (2025) distinguish hands-on "primary" care, which falls as children age, from continuous "on-call" care, which stays roughly constant and is hard to outsource; Ramey and Ramey (2010) document that parental childcare time rose sharply from the mid-1990s, especially among college-educated parents, and Guryan, Hurst, and Kearney (2008) show a positive education gradient in childcare time. Because more advantaged parents combine intensive parenting with jobs that grant short-run autonomy, they may face greater scope for caregiving to intrude on paid work — a tension between socioeconomic status and interruption risk that we document empirically below.

Evidence from Amuedo-Dorantes and Kimmel (2005) reinforces these cross-sectional

patterns. These authors show that the motherhood wage penalty in the United States is concentrated among women who do not delay fertility and is significantly reduced for college graduates — a finding consistent with human-capital and career-timing channels that our cross-sectional evidence on intensive-margin interruptions complements. These results also connect with the broader literature documenting that gender gaps in labour-market outcomes persist even after controlling for observable characteristics, pointing to normative and institutional mechanisms rather than differences in productivity (Blau & Kahn, 2017; Cortés & Pan, 2023).

2.2 Measuring Interruptions: Time Use, Sequence Analysis, and Event History

Time-use diaries are the natural instrument for studying daily caregiving, but they have typically been analysed at a resolution too coarse to see interruptions. Diaries record every activity of the day in sequence, yet most economic applications aggregate them into daily or hourly totals (Giménez-Nadal & Molina, 2022; Giménez-Nadal, Molina, & Velilla, 2020), and even sequence-aware studies of work effort (Giménez-Nadal & Sevilla, 2024) stop short of modelling sub-hourly transitions between work and care. This aggregation averages away exactly the high-frequency switches that constitute an interruption, leaving the within-day margin empirically invisible.

Recovering that margin calls for two tools: a way to represent the ordered sequence of daily activities, and a way to model the time until a target event. Social sequence analysis (Abbott, 1995; Cornwell, 2015), imported into the social sciences from biology, treats each diary as an ordered string of states and has been used to study careers, life courses, and daily activity patterns; event-history (survival) analysis models the time elapsed until an event and is a workhorse of labour economics, where it underpins the study of unemployment durations (Lancaster, 1979; van den Berg, 2001) and has been applied to long-run transitions such as maternal employment interruptions (Kleven, Landais, & Søgaard, 2019) and family formation (Persson & Rossin-Slater, 2024). Our contribution is to bring this machinery down to the intraday scale, modelling minutes-to-interruption rather than months-to-re-employment.

The closest precedent to our study, and a central reference for it, is the demonstration that work interruptions themselves carry a wage cost. Analysing two million tasks on an online platform, Adams-Prassl et al. (2025) find that women earn about 20 per cent less per hour despite no gender difference in task selection or experience, and that this gap is driven by women interrupting their paid work more often — with the penalty concentrated among workers with children. This is the clearest existing evidence that the act of interrupting work, rather than differences in human capital, can generate a gender gap in pay, and it directly motivates our question: if interruptions are costly and fall disproportionately on mothers, then understanding what triggers them within the workday is a first-order issue.

We build on this insight but differ in setting and in how the event is defined. Adams-Prassl et al. (2025) observe a single online labour market and identify interruptions as gaps of inactivity between tasks. We instead use nationally representative time-use diaries across eight countries and define an interruption by the *content* of the intervening activity — it must be childcare — separating cause from mere idleness, and we split *pure* from *mixed* episodes

and treat them as competing risks. This makes the typology an object of estimation rather than a description, and shifts attention from the documented *consequences* of interruptions to their real-time *risk*.

3 Data and Variables

3.1 Data Sources and Sample

Our primary data are the American Time Use Survey (ATUS) for 2003–2023, the benchmark source for time allocation in the United States. Administered by the U.S. Census Bureau for the Bureau of Labor Statistics, the ATUS draws respondents from households exiting the Current Population Survey and has one person aged 15 or older complete a 24-hour diary recording every activity with its duration, location, and co-presence. From 245,139 initial observations we retain a final analytical sample of 36,339 working parents after five restrictions: (i) employed; (ii) at least one child under 18; (iii) at least 60 minutes of paid work on the diary day; (iv) a high-quality diary; and (v) interruptions longer than 120 minutes recoded as censored rather than dropped. Survey weights (`ocombwt`) and household identifiers (`hldid`) are retained throughout for representative inference and clustered standard errors. Restriction (iii) is consequential: by conditioning on having worked that day, the sample isolates the *intensive* within-day margin and necessarily abstracts from the extensive participation margin, so our estimates describe interruption risk *conditional on working* rather than the full labour-supply response to children.

To gauge external validity we extend the analysis to eight countries in the Multinational Time Use Study (MTUS, 2007 onward): Armenia, Belgium, Canada, Finland, France, Spain, the United Kingdom, and the United States. The MTUS harmonises activity codes, demographics, and time formats across national surveys, and we identify respondents through a four-part key (country × survey × household × person); the pooled sample comprises 42,344 parents and 1,915 interruption events. Spanning welfare regimes that differ markedly in childcare provision and leave policy, this design lets us test whether the intraday gender gap is a US-specific artefact or a structural regularity.

3.2 The Interruption Event and the Covariates

The dependent variable is the time, measured from the start of paid work, until the first work interruption due to childcare — an episode of childcare that begins after the workday has commenced and is followed by a return to paid work. We distinguish *pure* interruptions, in which the intervening period is exclusively childcare, from *mixed* interruptions, which combine childcare with leisure or personal care but explicitly exclude domestic work; the exact detection conditions appear in table 2. Treating these as competing events (section 5.2) lets us ask not only whether mothers and fathers differ in interruption risk but whether any difference operates through one type of interruption or both.

The covariates capture the household and socioeconomic context that theory flags as relevant. To avoid the collinearity between single parenthood and spousal employment, we combine them into a single `familyStatus` variable with five mutually exclusive categories

(single parent; and, in couples, spouse not in paid work, employed full-time, employed part-time, or with employment status N/A), taking "spouse not in paid work" as the reference. The remaining covariates — gender, age, number of children, household size, age of the youngest child, urban residence, income quartile, education, citizenship, cohabitation, and a midweek-diary indicator — are summarised in table 1. Folding parenthood and spousal status into one variable keeps the hazard-model coefficients interpretable and avoids attributing a single underlying constraint to two correlated regressors.

4 Methodology

Our empirical strategy has three components: a detection step that turns diaries into events, a survival-analysis core that models the timing of those events, and a machine-learning layer that stress-tests the functional form and the ranking of predictors. We describe each in turn, with attention to how the resulting quantities should be read.

4.1 Detecting Interruptions with Sequence Analysis

We first translate each diary into a sequence and search it for the signature of an interruption. Every diary day is discretised into 144 intervals of ten minutes, and each interval is assigned one of five states — Personal Care, Paid Work, Domestic Work, Leisure, or Childcare. An interruption is the first occurrence of the ordered pattern Paid Work → non-Work → Paid Work in which the intervening block contains childcare, subject to the conditions in table 2; the choice of a ten-minute grid is innocuous, as results are stable across grids from 1 to 60 minutes (table 8). This representation borrows directly from social sequence analysis, which originated in the alignment of biological sequences and was brought into the social sciences by Abbott (1995) to study careers and life courses and is now applied to daily activity patterns (Cornwell, 2015). The payoff is a clean mapping from diary to data: the first qualifying pattern defines the *event*, and the number of minutes from the start of paid work to that pattern defines the *survival time* that the models below take as their outcome.

4.2 Modelling the Timing of Interruptions: Survival Analysis

Survival (event-history) analysis is the appropriate framework because our outcome is a *time until* an event subject to censoring: many workdays end without any interruption, so the only information they carry is that no interruption had occurred by the end of the observed work period. We begin non-parametrically with the Kaplan–Meier estimator (Kaplan & Meier, 1958), developed for clinical trials and now ubiquitous wherever durations are studied, which estimates the survivor function $S(t) = \Pr(T > t)$ — here, the probability that a parent reaches minute t of the workday without an interruption. A Kaplan–Meier curve is read from left to right as the share of parents still "surviving" (uninterrupted); two curves that pull apart indicate a group difference, which the log-rank test (Mantel, 1966) evaluates formally. The visible separation of the male and female curves, and the way the gap widens over the day, is itself the first sign that the proportional-hazards assumption introduced below may fail.

The core estimates come from the Cox proportional-hazards model (Cox, 1972), which writes the hazard — the instantaneous risk of being interrupted at time t, given survival to t — as

$$\lambda(t \mid \mathbf{x}) = \lambda_0(t)\, \exp(\boldsymbol{\beta}'\mathbf{x}), \tag{1}$$

where the baseline hazard $\lambda_0(t)$ is left unspecified (the model is semi-parametric) and is estimated only up to the covariate shift $\exp(\boldsymbol{\beta}'\mathbf{x})$. The Cox model is the standard tool for duration data in biostatistics and a workhorse of labour economics, where the same hazard framework underpins the analysis of unemployment spells and job tenure (Lancaster, 1979; van den Berg, 2001). Its coefficients are read as *hazard ratios*, $\mathrm{HR}_k = \exp(\beta_k)$: an HR of one means no association, an HR below one means lower instantaneous risk (and hence longer expected time to interruption), and an HR above one the reverse. Concretely, our central estimate of $\mathrm{HR} = 0.388$ for fathers means that, at any instant of the workday and holding the other covariates fixed, a father's risk of being interrupted is about 61 per cent lower than an otherwise identical mother's; the hazard ratio is a relative *rate*, not a probability or a ratio of durations, and does not by itself describe how the gap evolves over the day.

Two features of the data require adjustments to the Cox model. First, the ATUS is a complex survey with unequal selection probabilities and one diary per household, so we weight the partial likelihood by `ocombwt` and compute sandwich-robust standard errors clustered at the household level (Lin & Wei, 1989); ignoring the within-household correlation induced by shared norms and schedules would understate the standard errors and overstate significance. Second, the proportional-hazards assumption — that each HR is constant over the day — is itself testable. We inspect the Schoenfeld residuals, which should show no trend against time under proportionality; a non-zero slope signals a time-varying effect (fig. 2). Because the global test rejects proportionality decisively, we adopt a *stratified* Cox model that lets the baseline hazard differ freely across midweek and weekend diaries while estimating common covariate effects within strata, and treat it as our preferred specification. A stratified coefficient is interpreted exactly as before, but only *within* a stratum, with no cross-stratum comparison of baseline risk implied.

Finally, because pure and mixed interruptions are mutually exclusive ways the workday can be interrupted, we treat them as *competing risks* rather than collapsing them. We estimate cause-specific Cox models — in which the competing event is treated as censored — for each interruption type, and, as a check, a Fine–Gray subdistribution-hazard model (Fine & Gray, 1999) for pure interruptions. The distinction matters for interpretation: a cause-specific hazard ratio describes the *rate* of a given interruption type among parents still at work and answers an etiologic question (does gender act through this channel?), whereas the Fine–Gray subdistribution hazard describes the effect on the *cumulative incidence* of that type and answers a prognostic question (does gender change the eventual share experiencing it?). Reporting both guards against attributing the gender gap to one interruption type when it in fact runs through both.

4.3 Machine-Learning Benchmarks and Their Interpretation

Alongside the Cox models we estimate two machine-learning survival models. The motivation is diagnostic rather than causal: machine learning excels at flexible prediction and at letting

the data choose the functional form, but it estimates no structural parameter (Mullainathan & Spiess, 2017), so we use it to ask whether a flexible learner can out-predict the semi-parametric Cox and whether it agrees on which covariates matter — not to identify effects.

The Random Survival Forest (RSF) extends the random forest to censored data (Ishwaran et al., 2008): it grows 500 trees on bootstrap samples (here with `nodesize` = 15 and `mtry` = 4), splitting each node on the covariate and threshold that maximise the log-rank separation between the resulting groups, and averages their predictions. Forest-based estimators of this kind have become standard tools for prediction and for heterogeneity analysis more broadly (Wager & Athey, 2018). Predictive accuracy is summarised by the out-of-bag concordance index (C-index), and variable importance by permutation VIMP, read as the increase in out-of-bag prediction error when a covariate's values are randomly shuffled: the larger the deterioration, the more the model relied on that covariate. The Gradient Boosting Survival (GBS) model (Ridgeway, 1999) instead builds the predictor additively, fitting a sequence of shallow trees to the gradient of the Cox partial-likelihood loss, with the number of boosting iterations (314) chosen by five-fold cross-validation to avoid over-fitting.

We compare all models on the C-index, the survival analogue of the area under the ROC curve: it is the probability that, for a randomly chosen comparable pair of parents, the model assigns higher risk to the one who is in fact interrupted sooner. A C-index of 0.5 is no better than chance and 1.0 is perfect discrimination, so a value of 0.72 means the model ranks the earlier-interrupter correctly in roughly 72 per cent of comparable pairs.

To look *inside* the machine-learning models we use two complementary tools. SHAP (SHapley Additive exPlanations) values (Lundberg & Lee, 2017; Štrumbelj & Kononenko, 2014), computed with the `fastshap` package (Greenwell, 2023) on a subsample of $n = 2{,}000$ with 50 Monte-Carlo repetitions, borrow the Shapley value from cooperative game theory to split each individual prediction into additive contributions from each covariate; the mean absolute SHAP value gives a global importance ranking, while the sign of an individual SHAP value shows whether a covariate pushed that prediction toward higher or lower risk, and the scatter of SHAP values against the covariate reveals non-linearity and heterogeneity. Accumulated Local Effects (ALE) plots (Apley & Zhu, 2020), computed with the `iml` package (Molnar, Casalicchio, & Bischl, 2018), trace the average effect of a single covariate on the prediction across its range by accumulating local changes within narrow windows of the data; unlike partial-dependence plots they remain valid when covariates are correlated, because they never evaluate the model at implausible covariate combinations. An ALE curve is read like a regression-adjusted dose response: its slope gives the direction and its curvature the shape of a covariate's marginal contribution.

4.4 Cross-National Analysis

On the MTUS sample we estimate two complementary specifications. A pooled Cox model with country fixed effects (United States as the reference), survey weights, and household-clustered standard errors holds constant cross-country differences in baseline interruption rates while estimating a common set of covariate effects, so that the pooled male hazard ratio reflects within-country variation. We then estimate a separate Cox model for each country with at least 20 events and display the male hazard ratios in a forest plot. The country fixed effects absorb level differences in how often interruptions occur, while the country-specific

models reveal whether the *gender* gap — the object of interest — holds in each national context taken on its own.

5 Results

5.1 Sample and Non-Parametric Evidence

The US analytical sample is broadly representative of working parents and confirms that interruptions are rare daily events. table 1 reports the descriptives for the 36,339 individuals (1,544 events): the average parent is about 40 years old, gender is balanced (48.97 per cent male), households average 1.85 children and just under four members, and over 69 per cent hold higher education. Interruptions occur on 4.17 per cent of diary days — 3.41 per cent pure and 0.76 per cent mixed, with the remaining 95.83 per cent censored — so that, although individually short, these events are common enough across a large sample to be modelled precisely.

Non-parametrically, the gender gap is already visible and clearly time-varying. The Kaplan–Meier survival curves diverge steadily over the workday, so that by 300 minutes from the start of work women's probability of remaining uninterrupted is roughly 12 percentage points below men's, and the log-rank test rejects equality decisively ($p < 0.001$); the daily activity profiles in fig. 1 show the underlying behavioural difference. Stratified curves also reveal that interruption risk is not monotone in socioeconomic status — education and income are associated with *higher* risk, child age with a steep early-life peak, and household size with a U-shape — a pattern that both foreshadows the positive socioeconomic gradient in interruption risk and signals the non-proportional hazards that motivate our preferred specification.

5.2 The Gender Gap and Its Robustness

Parental gender is the largest and most precisely estimated predictor of interruption risk. In the weighted, household-clustered Cox model (table 3), fathers face an instantaneous hazard 61 per cent below mothers' (HR = 0.389; 95% CI: 0.346–0.438; $p < 2 \times 10^{-16}$), essentially unchanged in the stratified specification ($\text{HR}_{\text{strat}} = 0.388$). Household and family structure matter as theory predicts — the number of children raises risk (HR = 1.465), each additional year of the youngest child's age lowers it (HR = 0.950), household size buffers it (HR = 0.782), and single parenthood (HR = 2.072) and dual-full-time couples (HR = 1.760) carry the highest risk — while, strikingly, higher education (HR = 1.962) and income predict *more*, not fewer, interruptions — a positive socioeconomic gradient consistent with intensive parenting norms (Guryan, Hurst, & Kearney, 2008; Ramey & Ramey, 2010) and the short-run autonomy of high-skilled jobs (Goldin, 2014). That gender dominates the economic covariates, and that the stratified concordance (0.720) edges the unstratified (0.713), together point to a gap rooted in who responds to children rather than in household resources.

The gender gap runs through both types of interruption, not one. Cause-specific Cox models (table 4) yield a male hazard ratio of 0.383 for pure interruptions (1,259 events) and 0.420 for mixed interruptions (285 events), against 0.389 in the pooled specification, and a

Fine–Gray subdistribution model for pure interruptions returns a closely comparable 0.386. Because the estimate is nearly invariant to whether the interruption is exclusively childcare or combined with other activities, the gap cannot be dismissed as an artefact of one narrowly defined event type.

The estimate is also robust to how long interruptions are admitted as events before being recoded as censored. Re-estimating the gender hazard ratio under thresholds of 90, 120 (baseline), and 150 minutes, and with no cap, yields 0.385, 0.389, 0.404, and 0.447 respectively (table 6). The gap narrows modestly as longer interruptions enter — men are relatively more represented among long absences — but remains large and highly significant throughout, confirming that the result is not an artefact of the 120-minute convention.

5.3 Machine-Learning Results

The flexible learners reproduce the substantive ordering of predictors without improving on it. The Random Survival Forest attains an out-of-bag C-index of 0.617 and ranks the age of the youngest child first (VIMP = 0.094; relative importance 100 per cent), parental gender second (0.075; 79.5 per cent), and family status third (0.028; 29.7 per cent); the full convergence and VIMP ranking appear in fig. 3. The SHAP analysis (fig. 4, fig. 5) places gender first on mean absolute contribution (1.124), ahead of child age (0.866), family status (0.689), number of children (0.613), respondent age (0.585), and education (0.469). The two importance measures disagree only on which of gender and child age ranks first — a difference that reflects what each metric targets — but agree that these two dominate, reinforcing the Cox conclusion through an entirely different modelling lens.

The shapes uncovered by the interpretability tools match the non-parametric evidence and the Cox signs. The ALE profiles (fig. 6) show a steep convex rise in risk over child ages 0–4 that flattens at school age, a large discrete drop for fathers, a U-shaped household-size effect, and positively sloped education and income profiles consistent with the socioeconomic gradient documented above; the SHAP scatters tell the same story at the individual level. These curves do not overturn any Cox result but make its non-linearities — which linear main effects cannot display — directly visible.

Crucially, the flexibility of machine learning buys no predictive advantage here. The Gradient Boosting model attains a C-index of 0.711 — statistically level with the weighted Cox (0.713) and just below the stratified Cox (0.720) — with gender and child age again accounting for about 42 per cent of total importance (21.7 and 19.7 per cent); the full ranking appears in table 5. That a flexible learner matches but does not beat the semi-parametric Cox implies a hazard structure that is approximately additive and proportional within strata, which is why we treat the Cox estimates as primary and the machine-learning layer as diagnostic.

5.4 Cross-National Evidence

The interruption rate varies across countries, but the gender gap does not disappear anywhere. Across the eight-country MTUS sample (table 7, fig. 7), event rates range from 3.20 per cent in Spain to 9.51 per cent in the United Kingdom, with the United States (4.17 per cent) below the cross-country mean and gender balance close to even except in Armenia. These

level differences set up the central test: whether the gendered pattern documented for the United States recurs once institutional context changes.

The pooled model confirms the gap but must be read with care. A pooled Cox model with country fixed effects (United States as reference, weights, household clustering) estimates a male hazard ratio of HR = 0.388 (95% CI: 0.334–0.451), but because the United States supplies roughly three quarters of the pooled sample this coefficient is mechanically close to the US-only estimate and is not independent cross-national validation. The country fixed effects themselves indicate sizeable differences in baseline interruption rates — highest in Finland (HR = 14.92 vs. US) and the United Kingdom (HR = 12.78) — though such large fixed effects may partly reflect how each national diary instrument records paid work, a measurement caveat we attach to their interpretation.

The decisive evidence is the set of country-specific models, and it is unambiguous. fig. 8 and table 9 report male hazard ratios from separate Cox models for countries with at least 20 events: 0.388 in the United States, 0.390 in France, 0.466 in Spain, 0.518 in Belgium, 0.642 in the United Kingdom, and 0.657 in Finland (not significant at the 5 per cent level, $n_{\text{events}} = 47$). Canada (101 events) could not be estimated owing to sparse cells after stratification. No country shows a male hazard ratio above one, so fathers are less likely than mothers to interrupt paid work for childcare in every national context observed, which is the strongest possible form of the universality claim given the data.

Country-specific Random Survival Forests reinforce this reading. fig. 9 shows OOB convergence and VIMP rankings for the six countries with at least 50 interruption events (United States, United Kingdom, France, Belgium, Canada, and Spain). In every country, child age and parental gender rank among the two most important predictors, with family structure third in most cases. The consistency of predictor ordering across welfare regimes and institutional contexts — each trained independently, on national samples — suggests that cross-national differences in interruption rates largely reflect variation in household composition rather than country-specific residual variation.

6 Discussion

The picture that emerges is consistent across methods and points to a norms-based account. The age of the youngest child and parental gender are the strongest predictors throughout, ahead of number of children, household composition, and family status, and well ahead of economic indicators such as income; the ordering is the same under Kaplan–Meier, the weighted Cox, the forest, and SHAP. The concordance ranking — stratified Cox (0.720) ≈ gradient boosting (0.711) ≈ unstratified Cox (0.713) > RSF (0.617) — indicates that the dominant relationships are approximately log-linear and proportional within strata, so the substantive interpretation rests on a model whose assumptions the data largely support.

6.1 Gender, Norms, and the Socioeconomic Gradient

The gender effect is not merely large but appears irreducible to other characteristics. Moving from an unweighted Cox (HR = 0.459) to the survey-weighted, clustered specification (HR = 0.388) widens the gap, because partners within a household share labour-market and

normative conditions whose correlation, once accounted for, sharpens the estimate; the competing-risks results show the gap is, moreover, of nearly identical size for pure and mixed interruptions. Because gender remains the top predictor in a non-parametric ensemble that already conditions on family composition and labour-market attachment, the association is consistent with a direct role of gender norms in determining who responds to children during the workday — though, the design being observational, we read this as an association rather than a causal effect.

The positive association between socioeconomic status and interruption risk is robust across specifications and admits three readings that our data cannot separate. Higher education and income predict *more* interruptions, which could reflect *demand* (advantaged parents practise more intensive parenting; Cuevas-Ruiz et al. 2025; Guryan, Hurst, and Kearney 2008; Ramey and Ramey 2010), *capacity* (they hold jobs combining long-run demands with short-run autonomy and blurred work–family boundaries; Bloom et al. 2015; Goldin 2014), or *measurement* (their interruptions are simply more visible in a diary). Under the first two readings the gradient reflects greater scope to absorb care during work rather than greater need; under the third it is partly an artefact of how interruptions are recorded — and an observational design cannot adjudicate between them, a limit we flag explicitly. Household structure adds a further layer: the U-shaped household-size effect (buffering at medium sizes, overload at very large ones) and the family-status ranking (single parents and dual-full-time couples most at risk) suggest that the dual-earner nuclear model, for all its benefits, is structurally fragile to daily care shocks.

6.2 Universality Across Countries

The cross-national results establish that the gap is a structural regularity, not a US peculiarity. Male hazard ratios lie below one in all eight countries, and the magnitude varies in a way broadly consistent with cross-national differences in childcare provision, paternal leave take-up, and occupational norms (Kleven, Landais, & Leite-Mariante, 2025): largest (around 0.39) in the United States and France and closest to one in Finland and the United Kingdom. Even in relatively egalitarian settings the intraday burden falls disproportionately on mothers, so countries where the ratio approaches one serve as existence proofs that the gap can be compressed while those near 0.39 mark the largest scope for reform. The near-zero importance of the country indicator in the multi-country forest reinforces the point: the mechanism appears structurally similar across welfare regimes even where its magnitude differs.

7 Conclusions

This paper documents a new, high-frequency margin of gender inequality. Using high-resolution time-use data and a battery of survival and machine-learning methods, we show that parental gender and the age of the youngest child are the dominant predictors of short-term, within-day work interruptions due to childcare, ahead of household composition and well ahead of income; fathers face an instantaneous interruption hazard about 61 per cent below mothers', a gap that survives stratification, alternative censoring thresholds, a competing-risks decomposition, and an entirely non-parametric ensemble. Because the gap

is the single largest and most robust feature of the data, it is unlikely to be a proxy for correlated household characteristics.

Our main contribution is to provide cross-sectional evidence, at a daily frequency, of a gender gap in work interruptions that is consistent with the mechanisms proposed to explain the child penalty. We do not provide causal evidence, and our design conditions on having worked on the diary day, so the extensive participation margin — which accounts for the largest part of the long-run penalty (Kleven, Landais, & Søgaard, 2019) — lies outside our frame. What we do document is that the intensive daily margin is large, normatively driven, and universal. It is plausible that minute-level disruptions accumulate — through forgone tasks, higher cognitive load, and constrained schedules — into the forgone promotions and reduced hours that constitute the long-run penalty, but our design documents the existence and correlates of the interruptions rather than their cumulation into later outcomes; that link is left for future work. The cross-national extension, with male hazard ratios below one in all eight countries (from 0.388 in the United States to 0.657 in Finland), shows that the phenomenon is a structural feature of post-industrial economies rather than a US-specific artefact, and provides external validity for the mechanisms documented in the child penalty literature (Amuedo-Dorantes & Kimmel, 2005; Cortés & Pan, 2023).

The findings carry implications at two levels, conditional on that causal chain being established. At the workplace, predictable and genuinely flexible scheduling, formal short-notice caregiving protocols that do not penalise progression, and employer-sponsored rapid-response care could lower the incidence and cost of unplanned caregiving events, provided uptake is gender-neutral. At the state level, investment in early childhood care, subsidised emergency provision, and support for local rapid-response services would reduce families' reliance on ad hoc solutions and narrow the inequality in interruption risk driven by job autonomy. The forest plot in fig. 8 offers a benchmark: countries where the male hazard ratio approaches one demonstrate that the gap is compressible.

Several limitations bound these conclusions and frame the agenda ahead. The design is observational, so the estimates are associations, not causal effects; most importantly, it conditions on having worked on the diary day and therefore speaks to the intensive within-day margin rather than the extensive participation margin that accounts for the largest part of the long-run penalty, so integrating the two margins is a priority. Time-use diaries may also underreport multitasking or informal care, and the visibility of interruptions may differ across occupations and telework arrangements; external validity beyond the eight countries is not assured; and quasi-experimental evaluation of scheduling and childcare policies would be needed to identify which measures most effectively narrow the gap.

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Table 1: Descriptive Statistics

	Mean	**Std. Dev.**
Work interruptions		
Pure interruptions	3.46%	—
Mixed interruptions	0.78%	—
Global interruptions	4.25%	—
Individual characteristics		
Age	39.73	9.52
Male	48.97%	—
Citizen	89.33%	—
Household structure		
Number of children (nChild)	1.85	0.92
Household size (hhldSize)	3.95	1.16
Age of youngest child (ageChild)	7.20	5.08
Cohabitants (cohab)	77.56%	—
Socioeconomic characteristics		
Income (lowest 25%)	11.24%	—
Income (middle 50%)	44.89%	—
Income (highest 25%)	43.87%	—
Secondary education or lower	8.75%	—
Completed secondary education	21.91%	—
Higher education	69.34%	—
Family status		
Single Parent	17.02%	—
In Couple (Spouse not in paid work)	17.54%	—
In Couple (Spouse employed full-time)	48.32%	—
In Couple (Spouse employed part-time)	11.71%	—
In Couple (Spouse employment N/A)	5.42%	—
Contextual variables		
Urban area	83.65%	—
Midweek	65.25%	—

Notes: $n = 36{,}339$ individuals; 1,544 interruption events. ATUS 2003–2023. All estimates use ATUS survey weights (`ocombwt`). Censored observations: 95.75%. Binary variables reported as percentages; continuous variables as means with standard deviations. Reference category for family status is "In Couple (Spouse not in paid work)".

Table 2: Algorithmic Conditions for Interruption Detection

Condition	Logic	Justification
1	Interruption occurs after the first state of "Paid Work"	Restricts attention to interruptions within an ongoing workday.
2	Activity immediately preceding interruption is "Paid Work"	Ensures direct transition from work to childcare.
3	Individual returns to "Paid Work" after the interruption	Confirms interruption is within, not at the end of, the workday.
4	(Pure only) Entire intermediate period is exclusively "Childcare"	Expresses the purity requirement; excludes mixed episodes with leisure or personal care.

Notes: Mixed interruptions relax Condition 4, allowing the intermediate period to combine Childcare with Leisure or Personal Care, while explicitly excluding Domestic Work.

Table 3: Cox Proportional-Hazards and RSF Estimates — United States 2003–2023

	Cox (weighted, clustered)		RSF		SHAP
Variable	**HR**	**HR (stratified)**	**VIMP**	**Rel. Imp.**	**Mean $\lvert\phi\rvert$**
Male	0.388*** (0.334–0.451)	0.387*** (0.338–0.443)	0.032	55.9%	1.138
Age	1.002 (0.992–1.013)	1.002 (0.992–1.013)	—	—	0.626
Midweek	1.136 (0.957–1.349)	*stratified*	0.025	43.8%	0.346
Household size	0.773** (0.658–0.907)	0.773** (0.659–0.908)	—	—	0.488
Number of children	1.515*** (1.278–1.796)	1.513*** (1.277–1.794)	—	—	0.683
Age of youngest child	0.952*** (0.935–0.970)	0.952*** (0.935–0.969)	0.058	**100%**	0.923
Urban area	1.083 (0.916–1.279)	1.081 (0.915–1.278)	—	—	0.124
Income (lowest 25%)	0.685† (0.461–1.020)	0.692† (0.464–1.030)	—	—	0.352
Income (middle 50%)	0.840* (0.720–0.980)	0.842* (0.721–0.984)	—	—	—
In Couple (FT spouse)	1.837*** (1.508–2.237)	1.832*** (1.504–2.231)			
In Couple (PT spouse)	1.702*** (1.321–2.191)	1.697*** (1.317–2.185)	0.029	49.9%	0.767
In Couple (Spouse N/A)	0.554† (0.293–1.048)	0.554† (0.293–1.047)	—	—	—
Single Parent	2.160*** (1.580–2.954)	2.162*** (1.583–2.951)	—	—	—
Completed sec. education	1.409† (0.970–2.046)	1.406† (0.968–2.044)	—	—	—
Higher education	1.912*** (1.374–2.660)	1.907*** (1.369–2.656)	0.009	15.3%	0.501
C-index	0.713	**0.720**	0.617		—
Events	1,544		1,544		—
PH test	$< 2 \times 10^{-16}$		—		—

Notes: ATUS 2003–2023; $n = 36{,}339$. Cox models estimated with `ocombwt` survey weights and household-clustered robust standard errors (Lin & Wei, 1989). The stratified Cox model allows the baseline hazard to vary by `midWeek` stratum. RSF: 500 trees, `nodesize` = 15, `mtry` = 4, survey weights. SHAP values: `fastshap`, n = 2,000, `nsim` = 50. Reference categories: male (female), income (highest 25%), educaGroup (secondary or lower), familyStatus (In Couple, Spouse not in paid work). 95% confidence intervals in parentheses. † $p < 0.10$; ∗ $p < 0.05$; ∗∗ $p < 0.01$; ∗∗∗ $p < 0.001$.

Table 4: Gender Hazard Ratio by Interruption Type — Competing Risks

Specification	HR (male)	Lower 95%	Upper 95%	Events
Pooled (pure + mixed)	0.389***	0.346	0.438	1,544
Cause-specific: pure	0.383***	0.330	0.444	1,259
Cause-specific: mixed	0.420***	0.327	0.539	285
Fine–Gray (subdistribution, pure)	0.386***	0.329	0.453	1,259

Notes: ATUS 2003–2023; $n = 36{,}339$. Cause-specific Cox models treat the competing event as censored; all specifications include the full covariate set of table 3, survey weights, and household-clustered SE. $***p < 0.001$.

Table 5: Model Comparison — Concordance Index

Model	C-index
Cox stratified by midWeek (preferred)	**0.720**
Gradient Boosting Survival (302 iterations, CV)	0.711
Cox (weighted, household-clustered SE)	0.713
Random Survival Forest (weighted, 500 trees)	0.617

Notes: ATUS 2003–2023; $n = 36{,}339$; 1,544 events. C-index (concordance): 0.5 = random; 1 = perfect discrimination.

Table 6: Sensitivity of the Gender Hazard Ratio to the Censoring Threshold

Threshold	HR (male)	Lower 95%	Upper 95%	Events
90 minutes	0.385***	0.339	0.436	1,371
120 minutes (baseline)	0.389***	0.346	0.438	1,544
150 minutes	0.404***	0.358	0.456	1,641
No cap	0.447***	0.400	0.501	1,916

Notes: ATUS 2003–2023. Weighted Cox with household-clustered SE; full covariate set. Interruptions longer than the threshold are recoded as censored. $***p < 0.001$.

Table 7: Country Distribution — Multi-Country Sample (MTUS 2007+)

Country	n	**Events**	**Event rate**	**% Female**
Armenia	405	17	4.20%	41.5%
Belgium	1,324	81	6.12%	49.7%
Canada	1,854	101	5.45%	50.9%
Finland	685	47	6.86%	50.7%
France	3,071	154	5.01%	48.6%
Spain	2,250	72	3.20%	46.0%
United Kingdom	1,440	137	9.51%	50.4%
United States	31,315	1,306	4.17%	50.9%
Total	**42,344**	**1,915**	**4.52%**	**50.4%**

Notes: MTUS 2007+. Event rate: share of diary-day observations recording at least one work interruption due to childcare. % Female: share of female respondents. United States data from ATUS 2003–2023.

Table 8: Sensitivity Analysis to the Time Interval

	60 min	**30 min**	**15 min**	**10 min**	**5 min**	**1 min**
Male	0.585***	0.537***	0.486***	0.459***	0.444***	0.433***
	(0.483–0.709)	(0.458–0.630)	(0.424–0.557)	(0.404–0.522)	(0.392–0.502)	(0.383–0.490)
Age	1.030***	1.009*	1.006	1.011***	1.009**	1.007
	(1.017–1.043)	(0.999–1.020)	(0.997–1.014)	(1.003–1.020)	(1.002–1.018)	(0.999–1.015)
Midweek	1.477***	1.409***	1.274***	1.247***	1.262***	1.200***
	(1.157–1.885)	(1.158–1.454)	(1.089–1.492)	(1.076–1.444)	(1.096–1.454)	(1.045–1.378)
Household size	0.661***	0.666***	0.669***	0.685***	0.702***	0.729***
	(0.539–0.810)	(0.563–0.788)	(0.582–0.769)	(0.603–0.779)	(0.623–0.792)	(0.649–0.819)
Number of children	1.569***	1.612***	1.570***	1.586***	1.537***	1.492***
	(1.257–1.959)	(1.342–1.936)	(1.349–1.828)	(1.379–1.825)	(1.347–1.755)	(1.311–1.697)
Age of youngest child	0.921***	0.946***	0.948***	0.942***	0.944***	0.946***
	(0.901–0.942)	(0.929–0.963)	(0.934–0.963)	(0.928–0.955)	(0.931–0.957)	(0.933–0.959)
Income (lowest 25%)	0.605**	0.540***	0.608***	0.687***	0.699**	0.648***
	(0.408–0.899)	(0.386–0.755)	(0.468–0.789)	(0.540–0.872)	(0.557–0.878)	(0.514–0.816)
Income (middle 50%)	0.662***	0.684***	0.684***	0.742***	0.776***	0.773***
	(0.541–0.811)	(0.580–0.806)	(0.595–0.786)	(0.652–0.844)	(0.686–0.878)	(0.684–0.874)
Higher education	1.590*	2.350***	1.529**	1.932***	2.101***	2.435***
	(0.970–2.604)	(1.449–3.811)	(1.097–2.130)	(1.380–2.704)	(1.509–2.923)	(1.713–3.461)
Single Parent	1.275	1.218	1.269*	1.365**	1.480***	1.506***
	(0.864–1.882)	(0.878–1.689)	(0.971–1.658)	(1.064–1.752)	(1.168–1.876)	(1.193–1.901)
Concordance	0.689	0.681	0.679	0.682	0.682	0.684
Events	537	788	1,121	1,281	1,393	1,413

Notes: ATUS 2010–2019 (baseline specification); n = 20,494. Hazard ratios with 95% confidence intervals in parentheses. $*p < 0.10$; $**$ $p < 0.05$; $***p < 0.01$.

Table 9: Country-Specific Male Hazard Ratios — Multi-Country Cox Models

Country	**HR (male)**	**Lower 95%**	**Upper 95%**	**Events**
United States	0.388***	0.334	0.451	1,306
France	0.390***	0.267	0.569	154
Spain	0.466**	0.256	0.846	72
Belgium	0.518**	0.299	0.898	81
United Kingdom	0.642**	0.449	0.917	137
Finland	0.657	0.347	1.242	47

Notes: Country-specific Cox models with survey weights and household-clustered SE, controlling for age, midWeek, hhldSize, nChild, ageChild, urban, income, and educaGroup. Armenia (17 events) is excluded for having fewer than 20 events. Canada (101 events) could not be estimated due to sparse income–education cells after stratification. $**$ $p < 0.05$; $***$ $p < 0.001$.

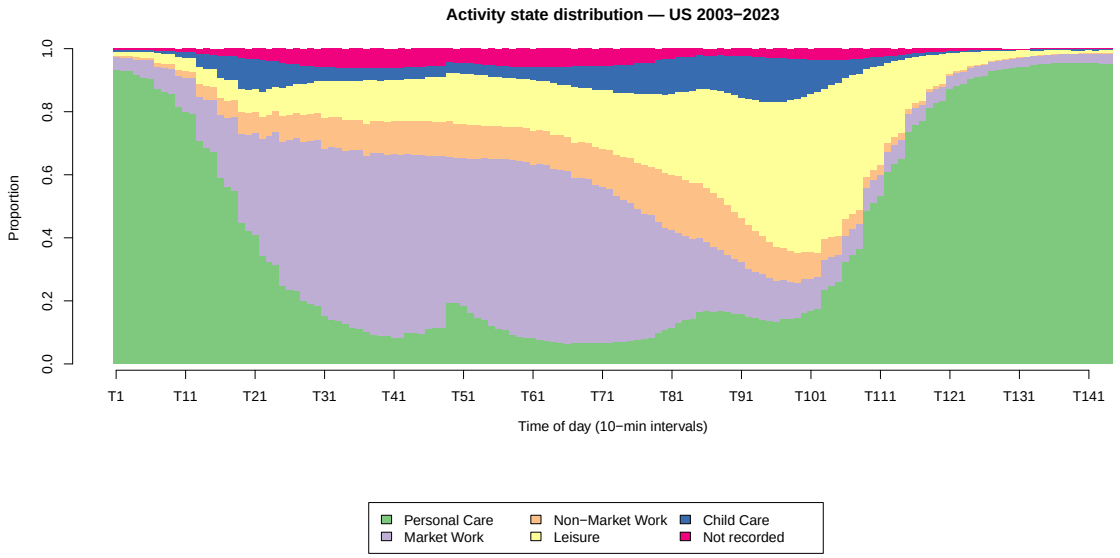


Figure 1: Distribution of daily activity states by gender (United States, 2003–2023)
Notes: Stacked area chart showing the share of working parents engaged in each activity state across the 24-hour day, averaged over all diary days (US sample, $n = 36{,}339$). Activity categories follow the MTUS harmonised coding.

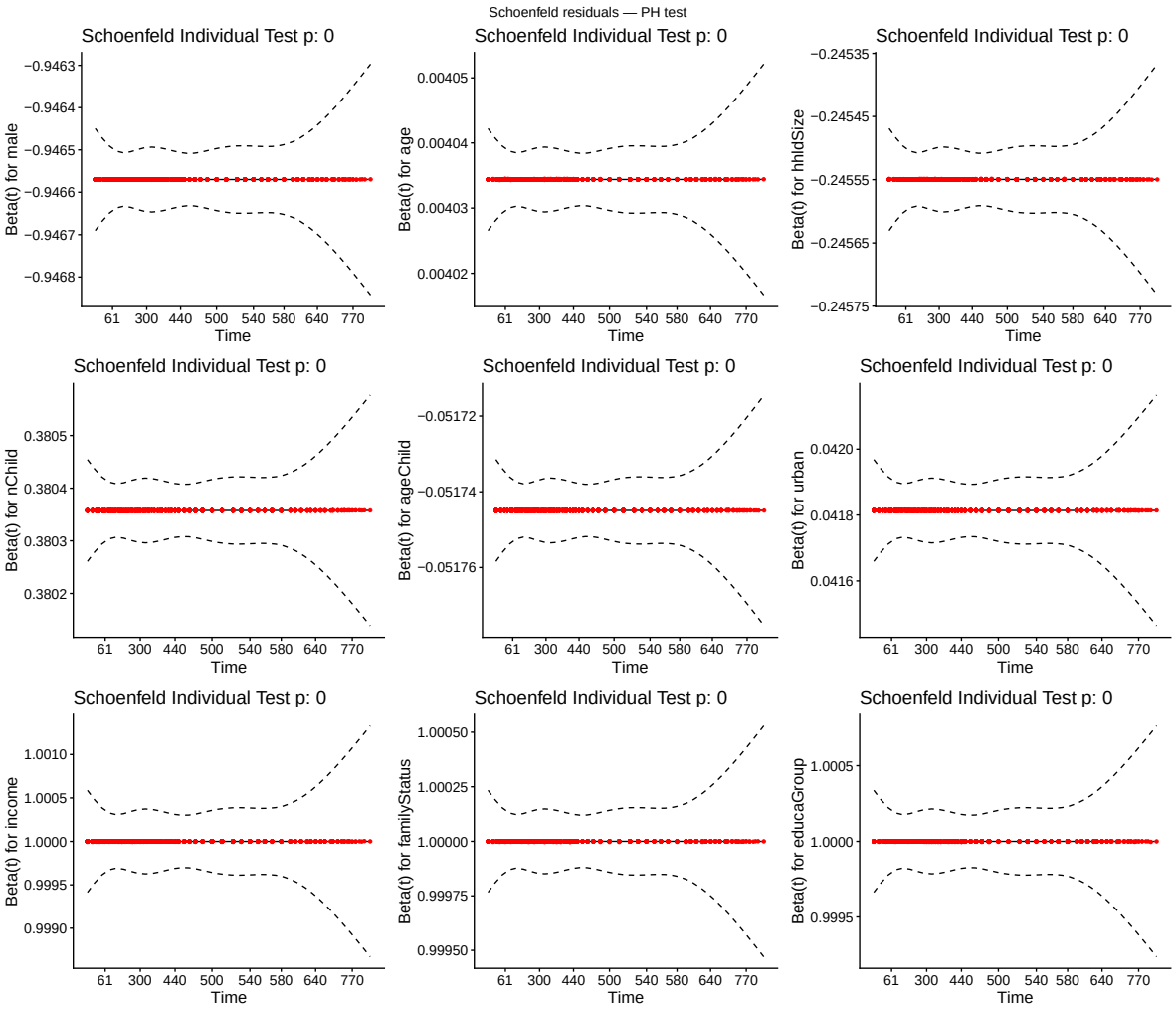


Figure 2: Schoenfeld residuals — global proportional-hazards test

Notes: Diagnostic plots of Schoenfeld residuals against time for each covariate in the Cox model. A non-zero slope indicates violation of the proportional-hazards assumption. Global test: $p < 2 \times 10^{-16}$.

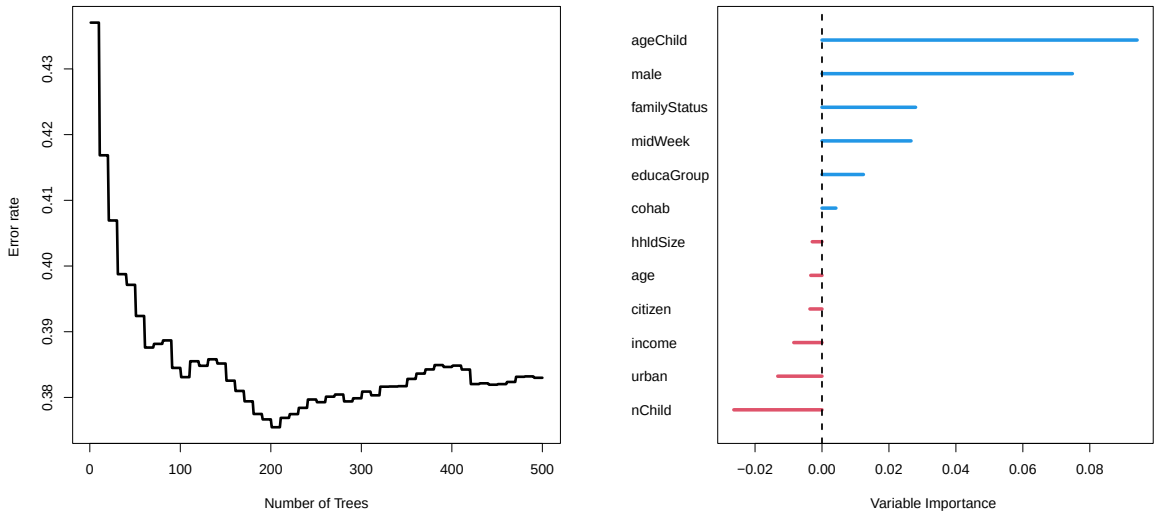


Figure 3: RSF: OOB convergence and variable importance (VIMP)
Notes: Left panel: OOB error rate against number of trees (convergence). Right panel: Permutation VIMP ranking. `ageChild` and `male` are the two most important predictors. RSF: 500 trees, `nodesize` = 15, `mtry` = 4, survey weights. ATUS 2003–2023.

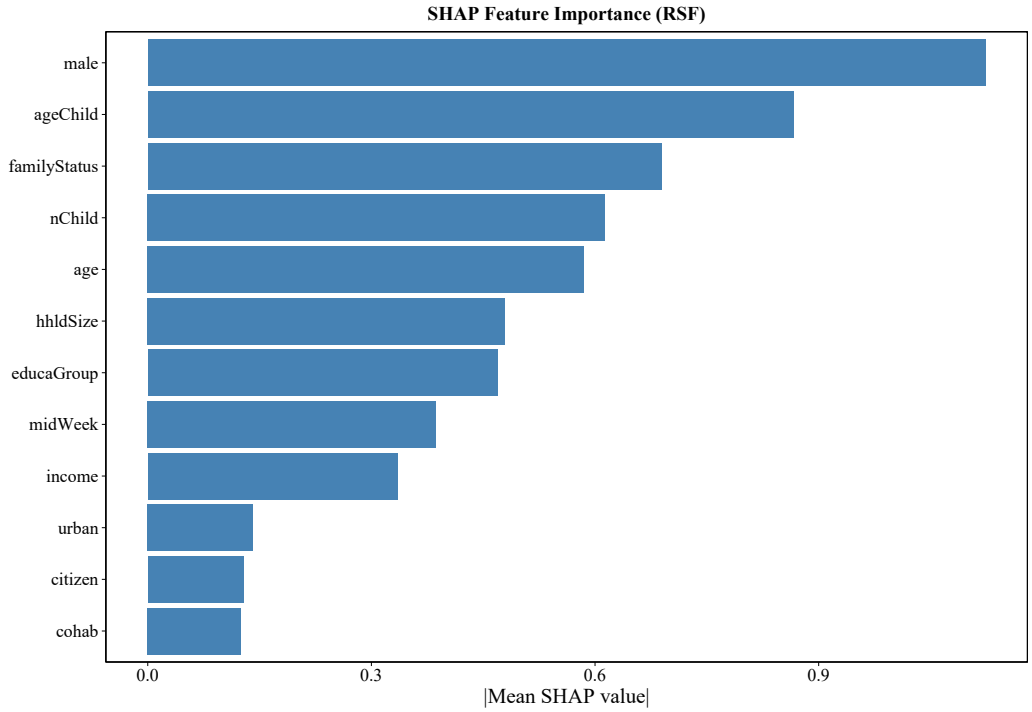
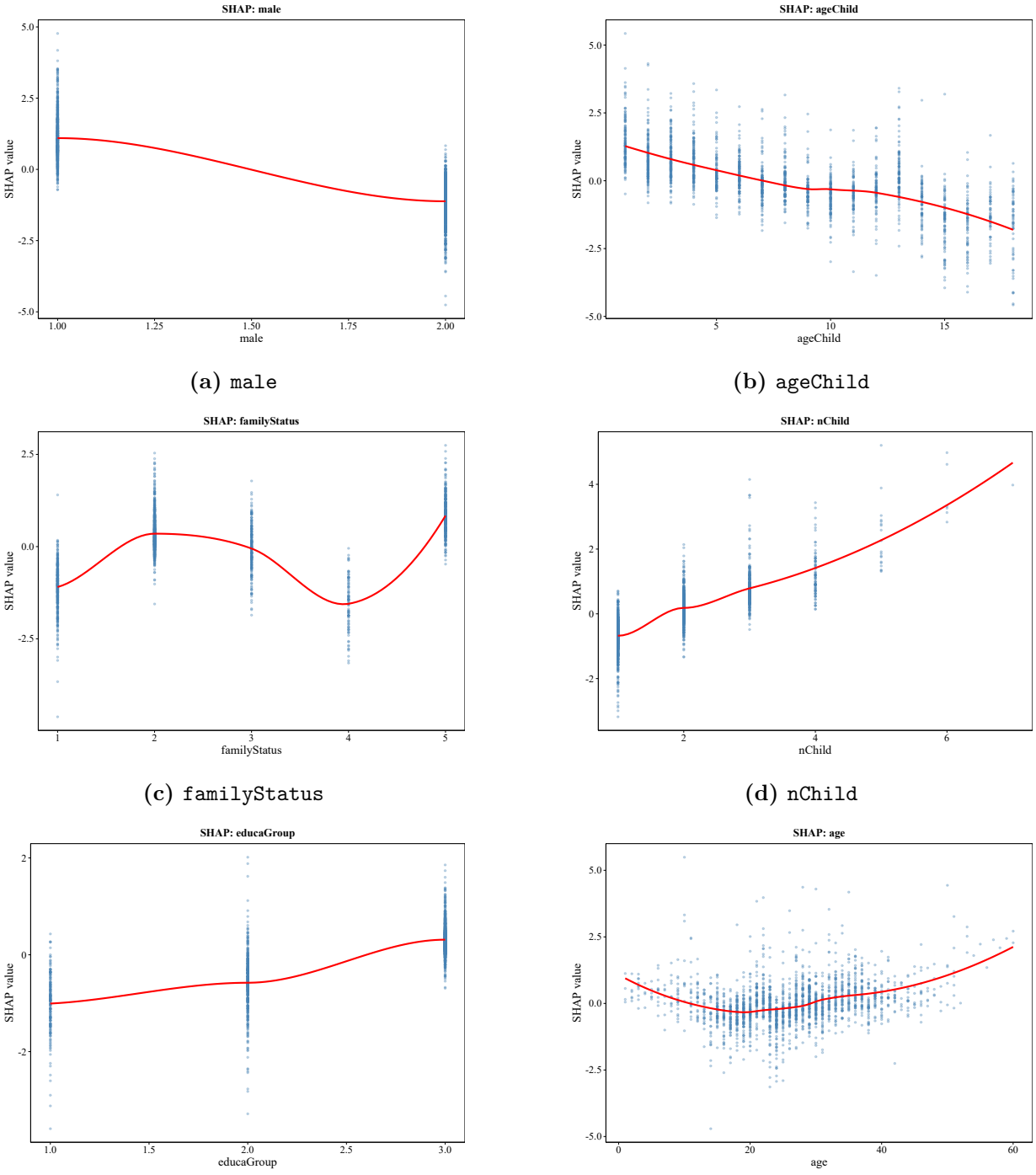


Figure 4: SHAP values: global variable importance
Notes: Mean absolute SHAP values (global importance ranking) from the RSF predictions. Computed with `fastshap` (n = 2,000, $nsim$ = 50). Positive SHAP values indicate higher predicted interruption risk. The six panels below show per-variable scatter plots for the top predictors.



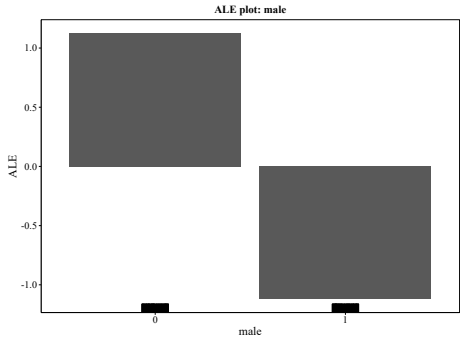
(a) `male` **(b)** `ageChild`

(c) `familyStatus` **(d)** `nChild`

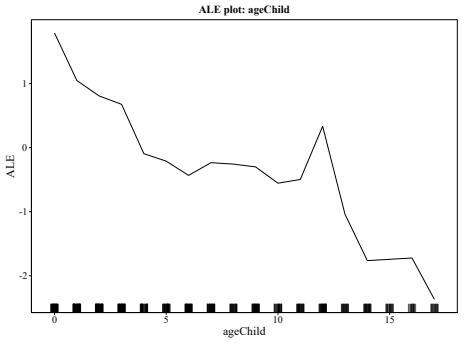
(e) `educaGroup` **(f)** `age`

Figure 5: SHAP scatter plots: top six predictors

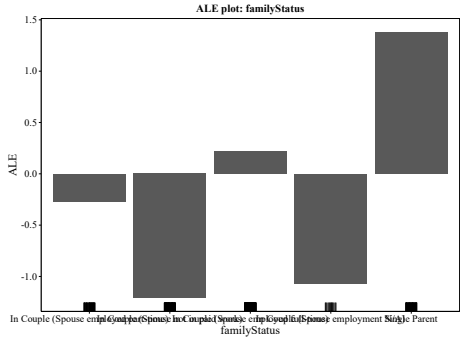
Notes: Each panel plots individual SHAP values against the feature value for the six most important predictors. A LOESS smoother (red) highlights the average marginal effect. Positive SHAP values indicate higher predicted interruption risk.



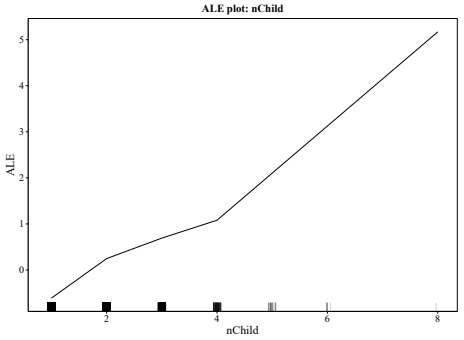
(a) `male`



(b) `ageChild`



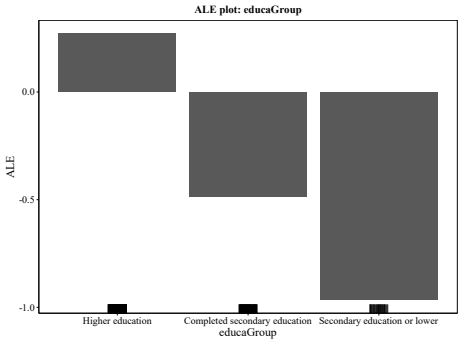
(c) `familyStatus`



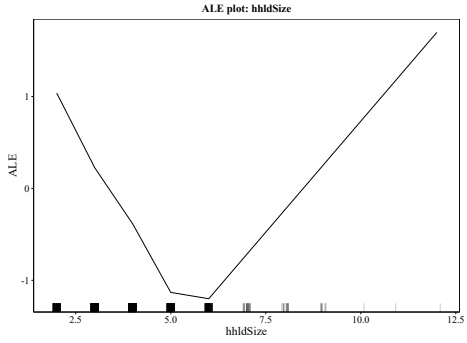
(d) `nChild`



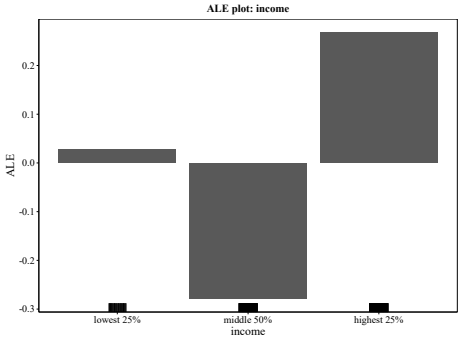
(e) `age`



(f) `educaGroup`



(g) `hhldSize`



(h) `income`

Figure 6: Accumulated Local Effects (ALE) plots — top eight predictors

Notes: ALE plots estimated on a subsample (n = 2,000) using the `iml` package. The y-axis represents the average change in predicted interruption risk attributable to each variable. ALE plots avoid distributional extrapolation under correlated predictors.

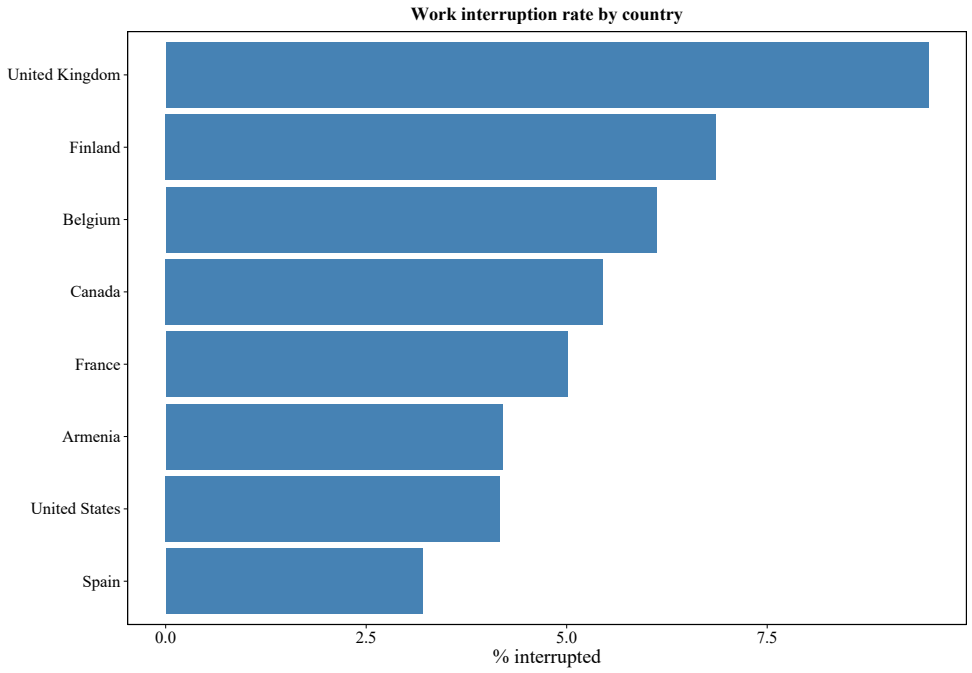


Figure 7: Interruption event rates by country

Notes: Bar chart of the share of diary days recording at least one work interruption due to childcare, by country. MTUS 2007+; $n = 42{,}344$.

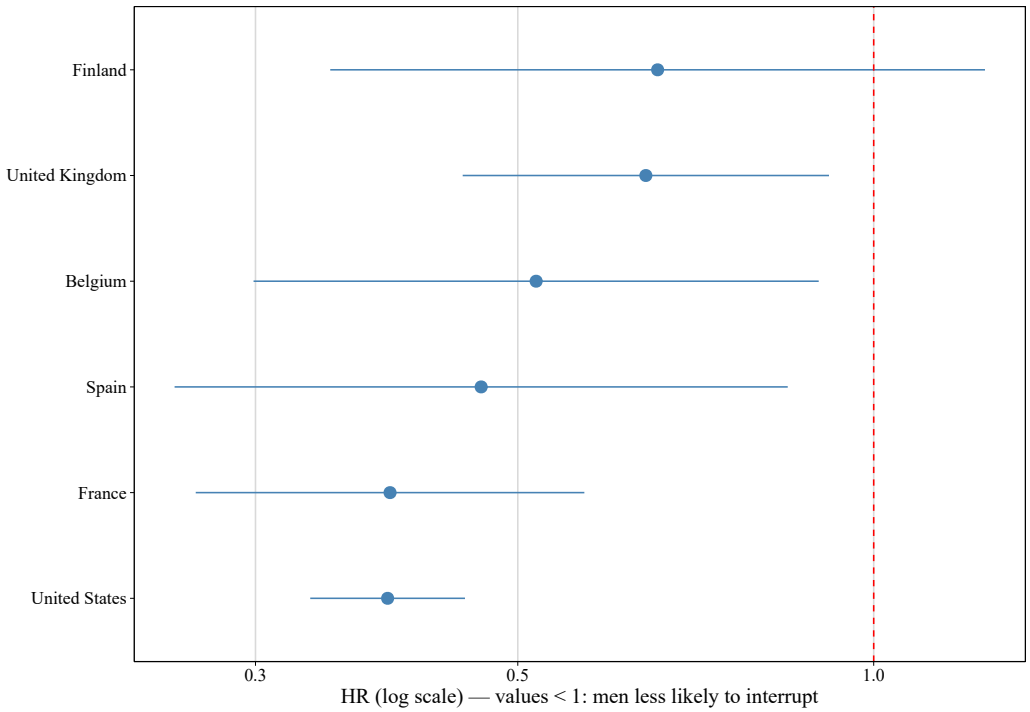
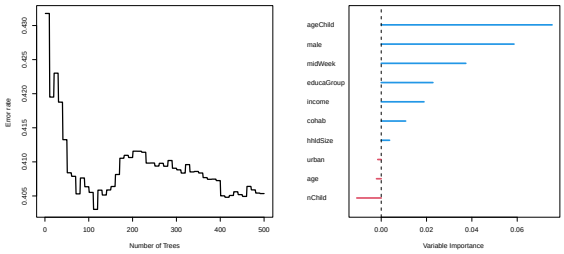
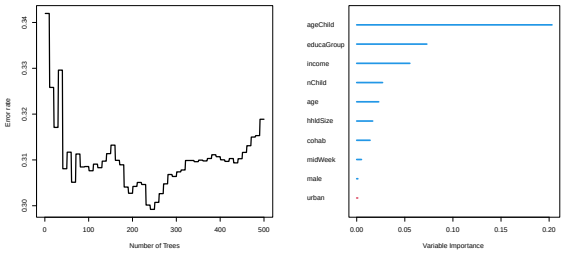


Figure 8: Forest plot: male gender hazard ratio by country

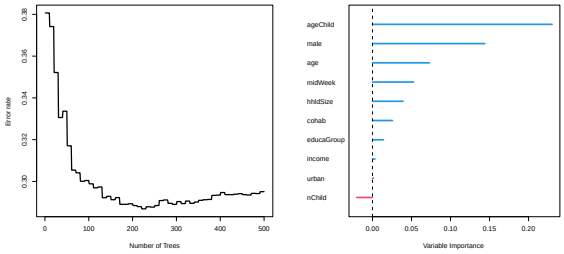
Notes: Hazard ratios for the male gender indicator from separate Cox models estimated for each country, controlling for age, household size, number of children, age of youngest child, urbanicity, income, and education (family status included where available). Survey weights and household-clustered robust standard errors. Error bars: 95% confidence intervals. Countries with fewer than 20 interruption events excluded. All point estimates lie below one. MTUS 2007+.



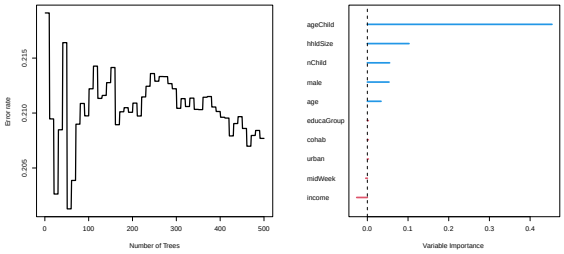
(a) United States



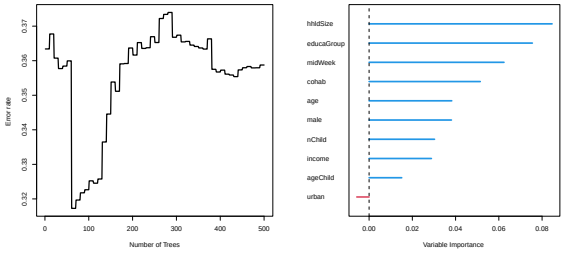
(b) United Kingdom



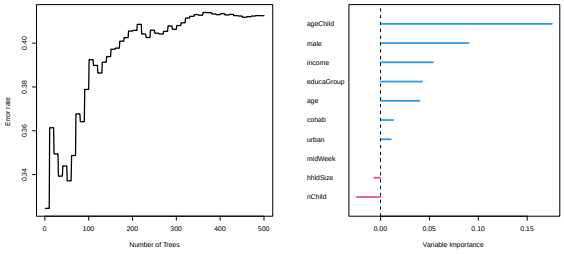
(c) France



(d) Belgium



(e) Canada



(f) Spain

Figure 9: Per-country RSF: OOB convergence and variable importance (VIMP)
Notes: Each panel shows OOB error convergence (left) and permutation VIMP ranking (right) for a country-specific RSF (500 trees, survey weights). Only countries with ≥ 50 interruption events are included. Finland (47 events) and Armenia (17 events) are excluded. MTUS 2007+.